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AERO60492 – Autonomous Mobile Robots

Lecture 11 – Rigid Body Dynamics 1

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Aims

- Understand rigid body dynamics modelling
- Newton-Euler modelling
- Derive generalised equations of motion

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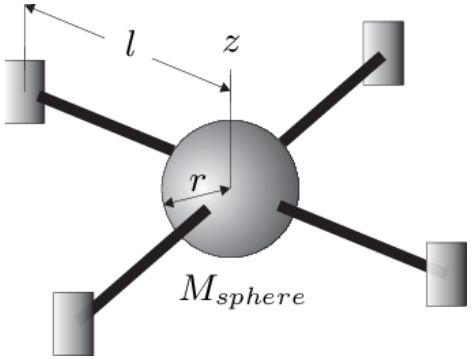
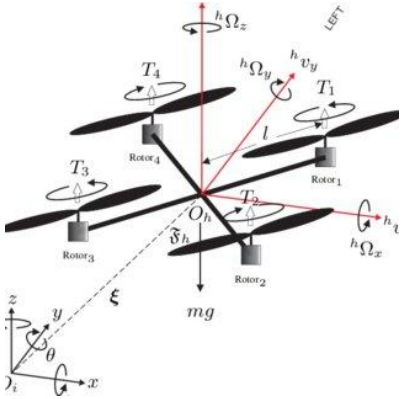
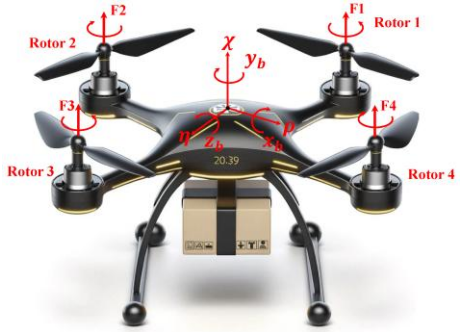
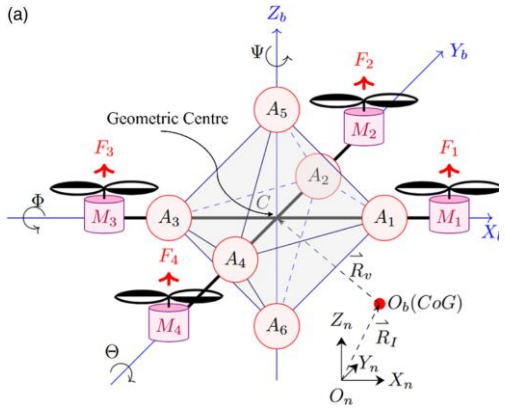
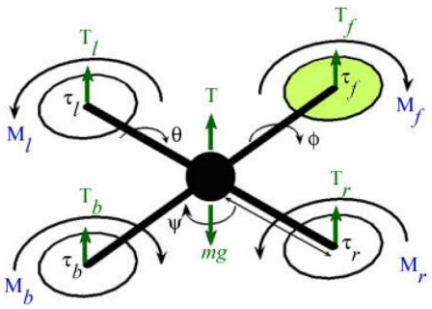


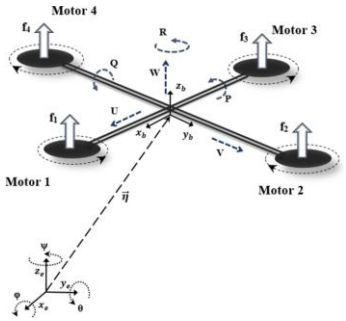


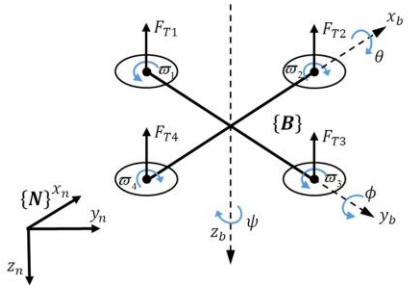


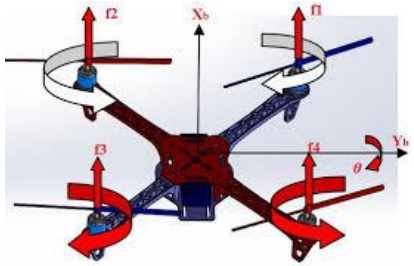


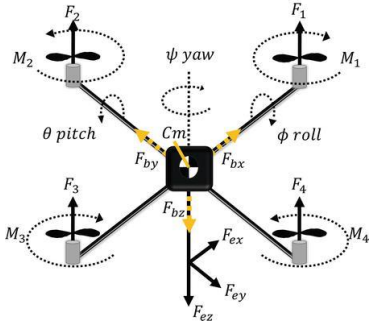


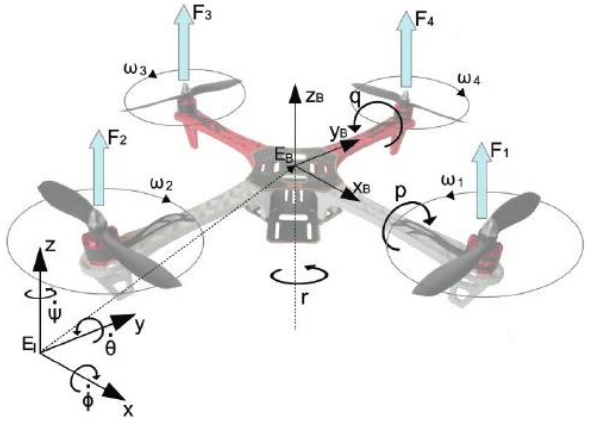
Common approach to modelling of rigid body dynamics!

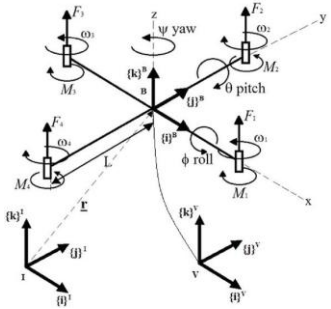


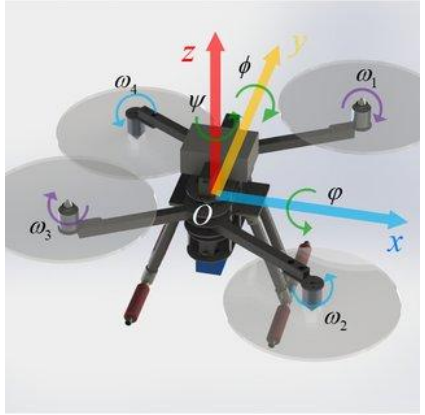


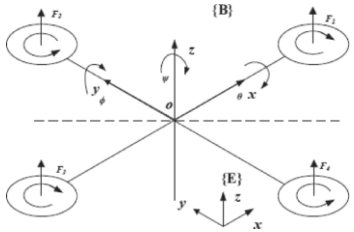


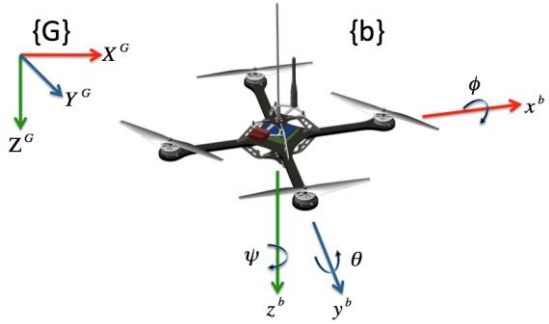


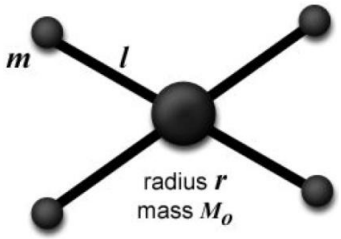


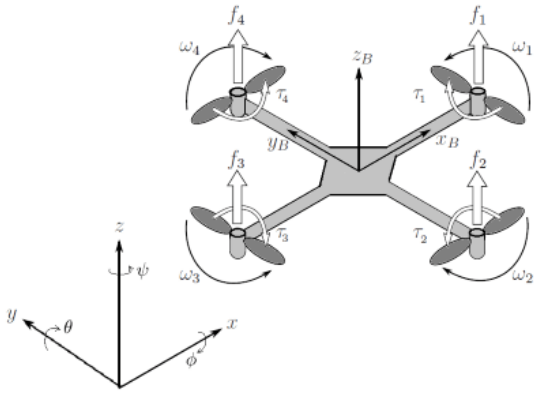












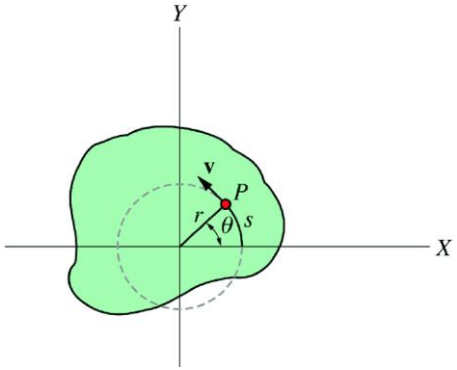
What is “rigid body dynamics”

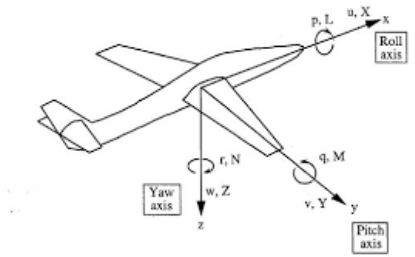
Movement of systems of interconnected bodies under the action of external forces

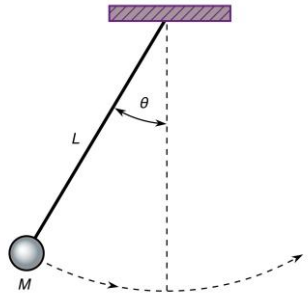
Rigid – the bodies are assumed connected by infinitely stiff structure 100% transmission of forces and torques between sub-components

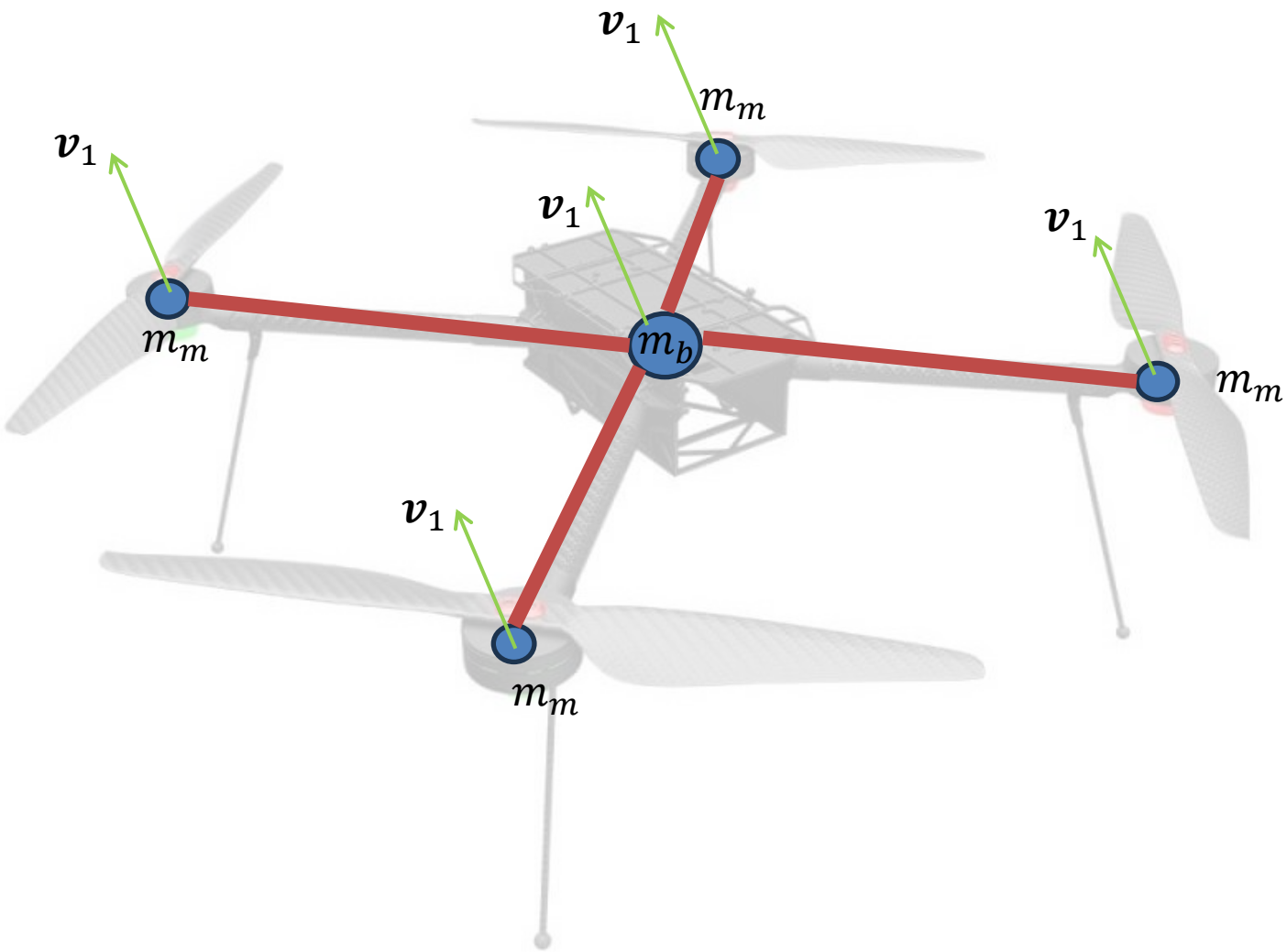
Dynamics – the science of forces and motion

Implies time is a dimension









$$m = \sum m_i$$

$$\boldsymbol{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix}$$

What is “rigid body dynamics”

Newton's second law expanded to:

- Three dimensions
- Translational and rotational
- Assuming constant mass and inertia

Translational $$F = ma$$

Force = mass X acceleration

Rotational $$T = I\dot{\omega}$$

Moment = moment of inertia X angular acceleration

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Reference Text

Equations and derivation based on:

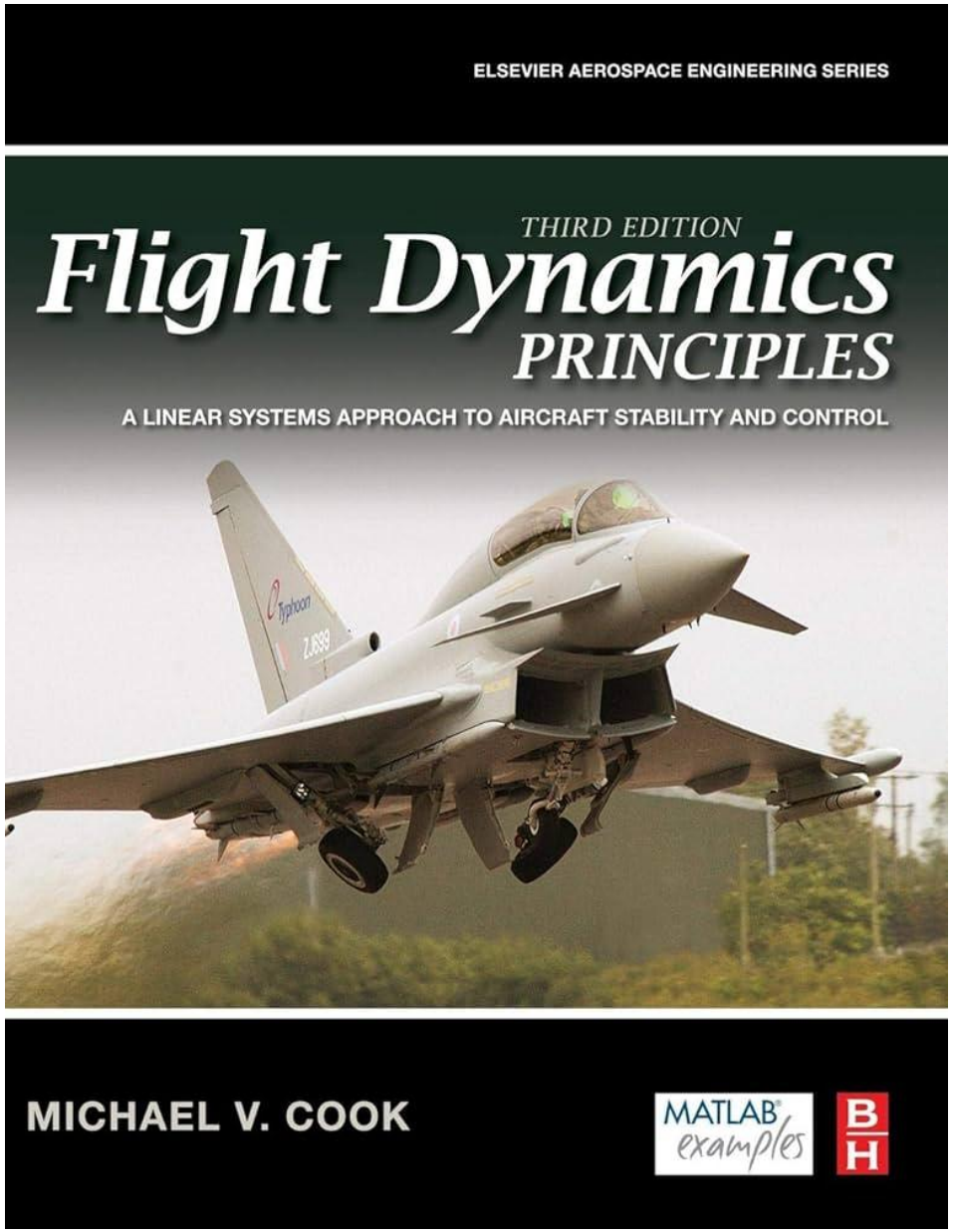
Flight Dynamics Principles

A Linear Systems Approach to Aircraft Stability and Control

Book • Third Edition • 2013

Michael V. Cook

Chapter 4: The Equations of Motion

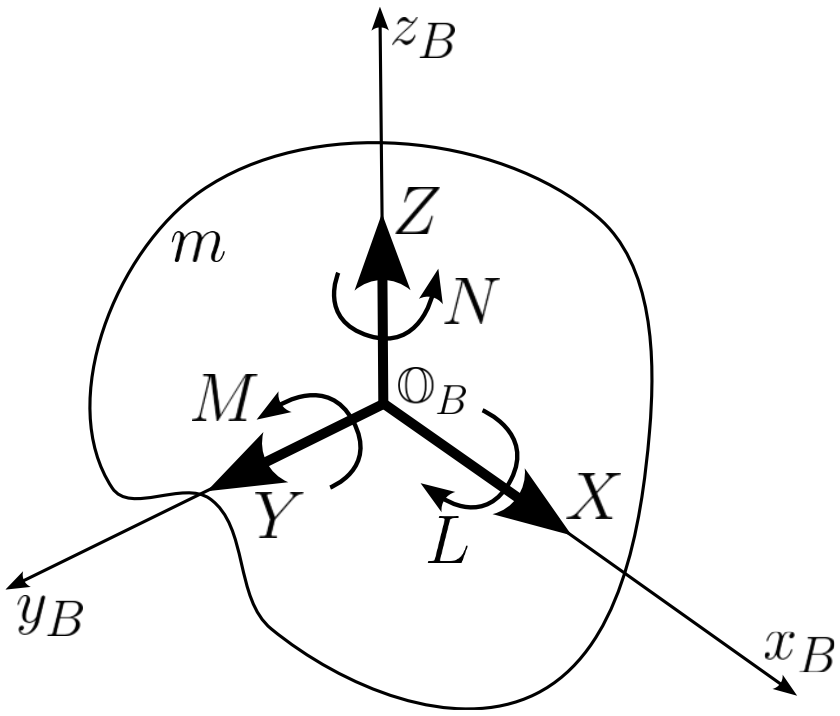


Forces and Moments

All forces and moments for a rigid body system, are assumed to act around a combined centre-of-gravity aligned to the body axes origin $\mathbb{O}_B$

All forces and moments are resolved into the body reference frame axes and coordinate system $(x, y, z)_B$

All forces and moments can be time evolving, but assumed instantaneous when calculating rigid body motion



$$\bar{F} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \qquad \bar{M} = \begin{bmatrix} L \\ M \\ N \end{bmatrix}$$

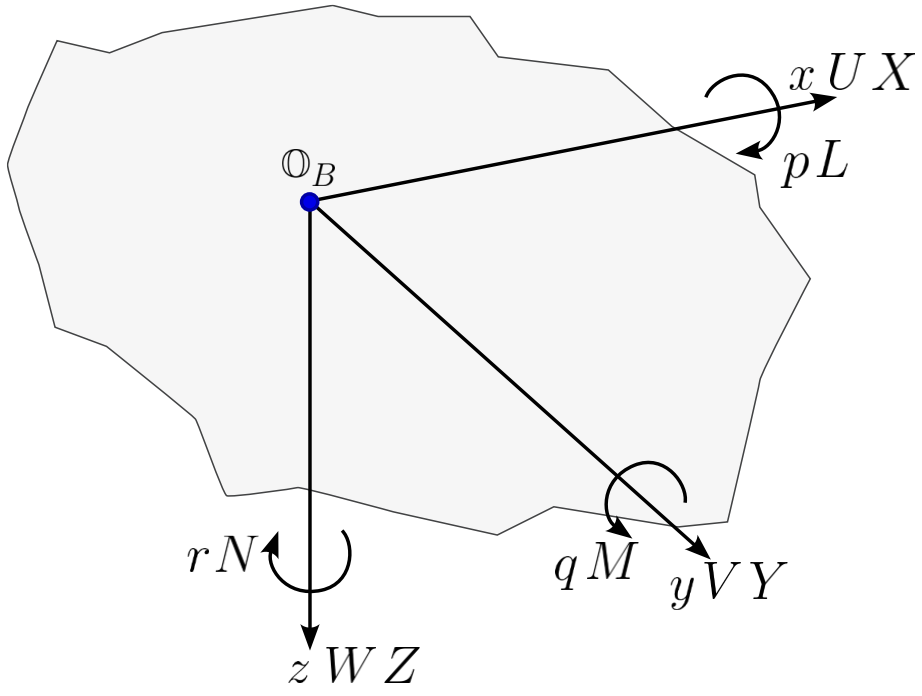
Force(s) vector Moment(s) vector

Generalised inertial acceleration

Define a **rigid body** in motion with respect to an external frame

It has translational force $\bar{F}$ and velocity $\bar{V}$ components **along** axes

It has rotational moments $\bar{M}$ and rates $\bar{\Omega}$ acting **around** axes



Forces and torques* relative to **inertial frame**

$$\bar{F} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \qquad \bar{M} = \begin{bmatrix} L \\ M \\ N \end{bmatrix}$$

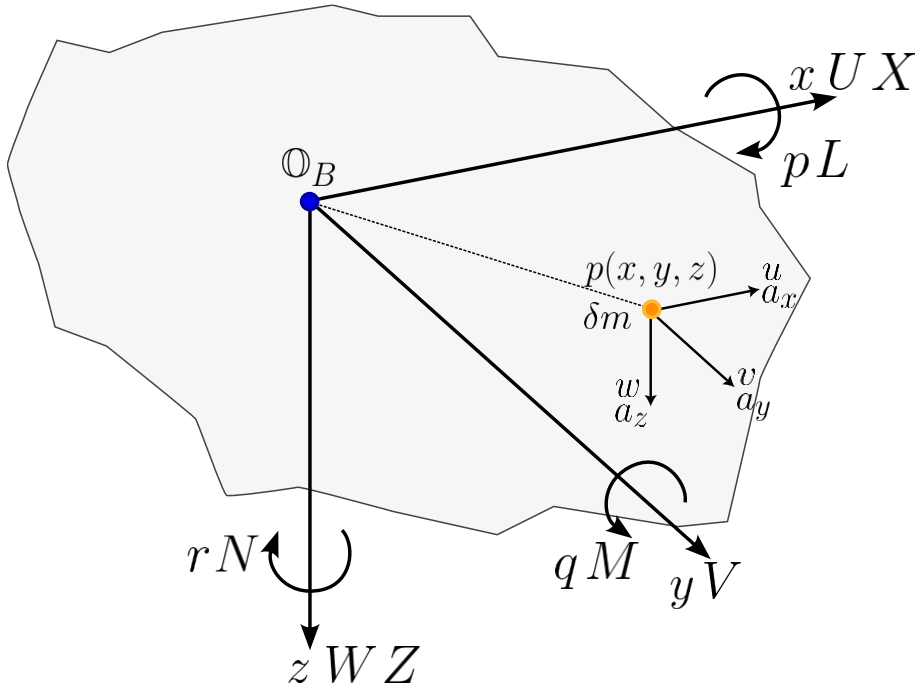
Velocities and rates* relative to **inertial frame**

$$\bar{V} = \begin{bmatrix} U \\ V \\ W \end{bmatrix} \qquad \bar{\Omega} = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

*positive rotation direction follows “curl right hand” rule

Generalised inertial acceleration

Define a point p within the body reference frame at coordinate (x, y, z)
Point has translational velocity (u, v, w) relative to the body frame
Point has translational acceleration (a_x, a_y, a_z) relative to the body frame
The point has an infinitesimal mass δm, which is used later



Position, velocity and acceleration relative to **body frame**

$$\bar{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

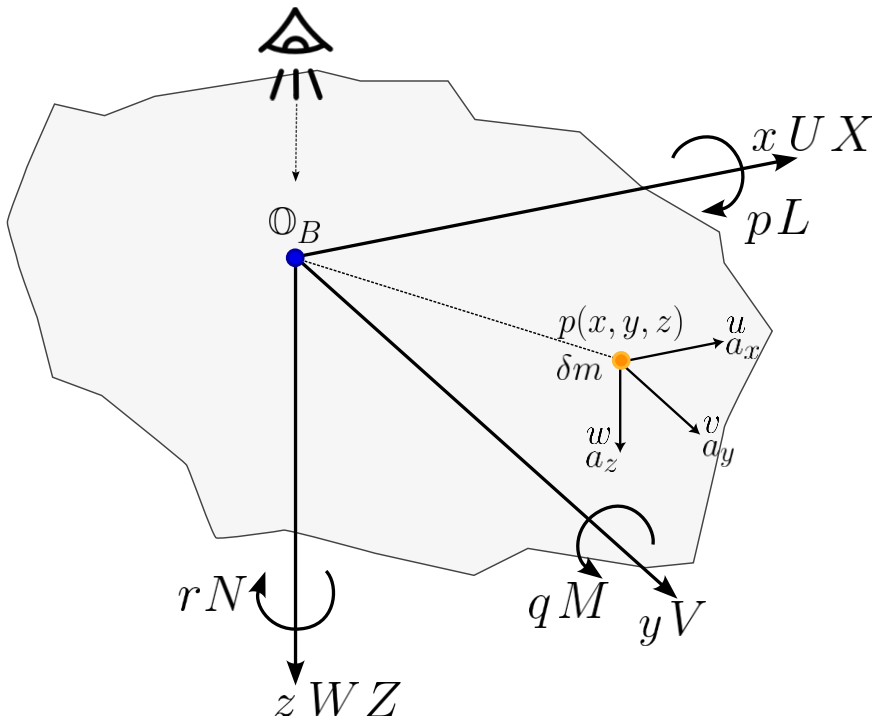
$$\bar{v} = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$\bar{a} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}$$

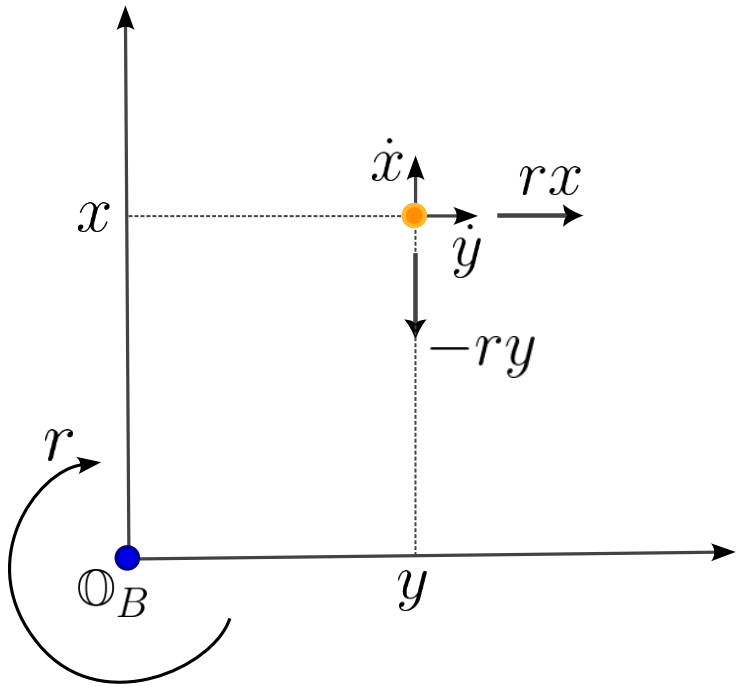
Generalised inertial acceleration

Sum all velocities acting at point p within body frame

- Direct velocity due to motion within the frame $(\dot{x}, \dot{y}, \dot{z})$
- Tangential velocity due to rotational motion (this is the application of "$v = \omega r$" in multiple axes)



Derive tangential velocity components by "viewing" along each axis and applying tangential velocity equation for a point at a radial distance from the centre of rotation

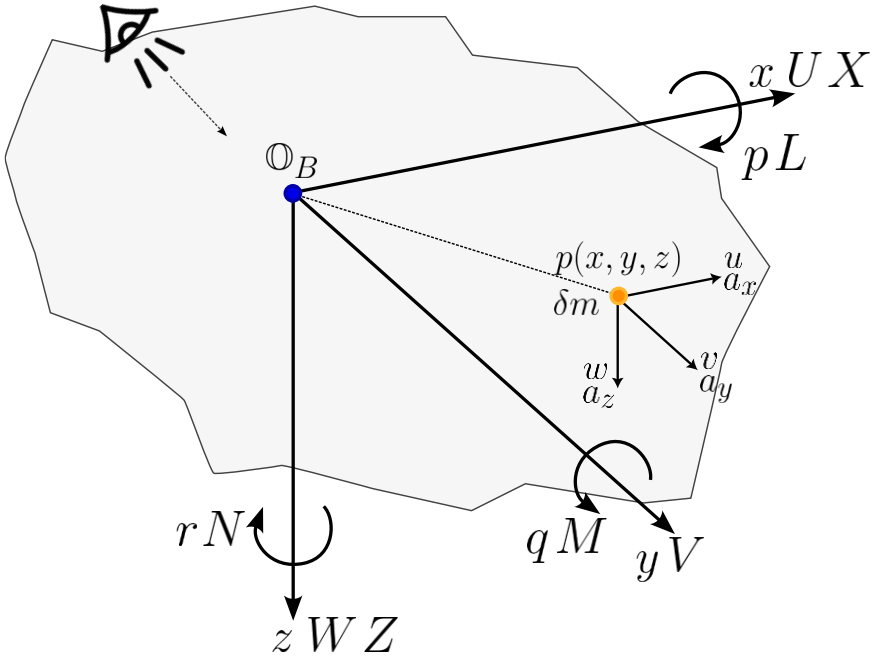


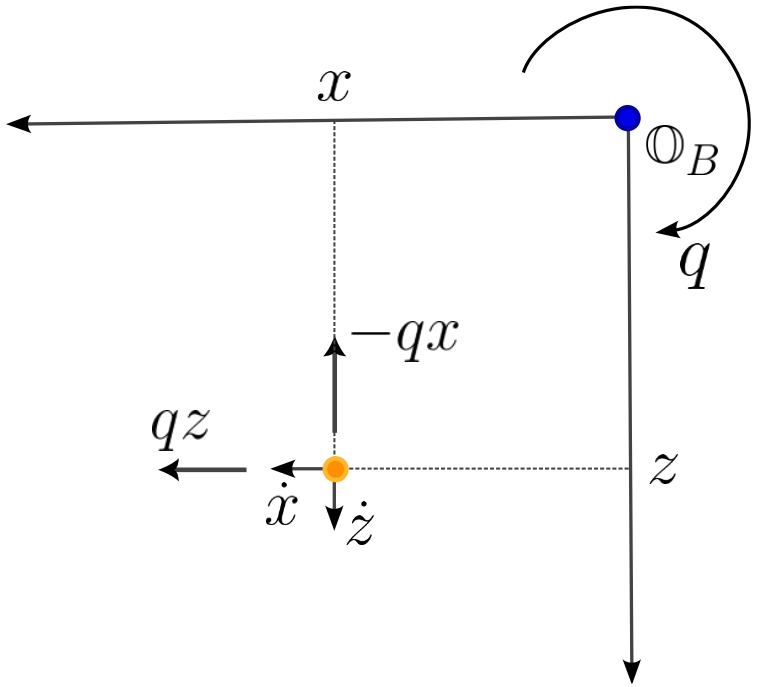
View looking along z axis

Generalised inertial acceleration

Here the axes are rotating only in the y axis at rate q

The point is offset from the centre of rotation by radius (distance) x and z in the other orthogonal axes



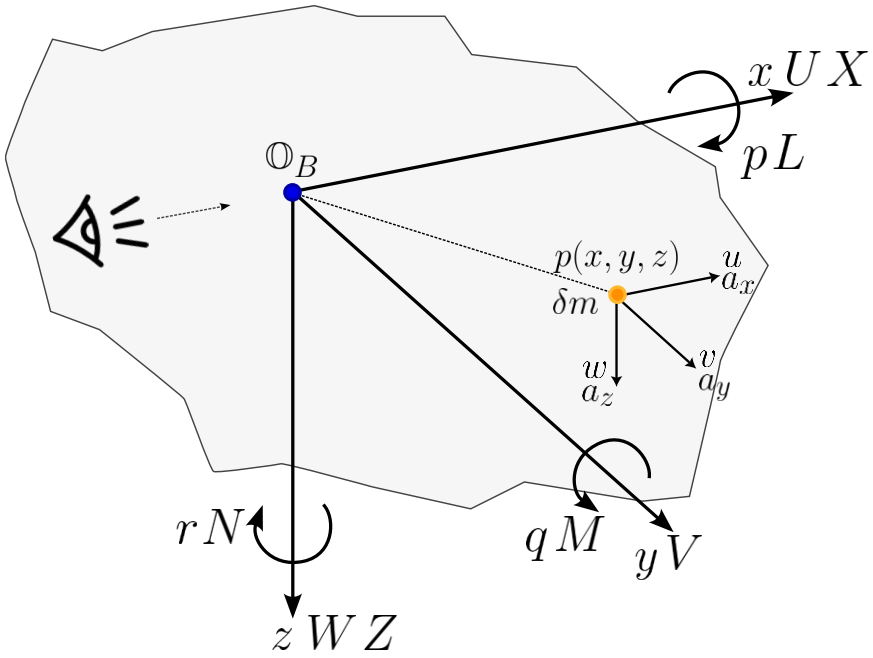


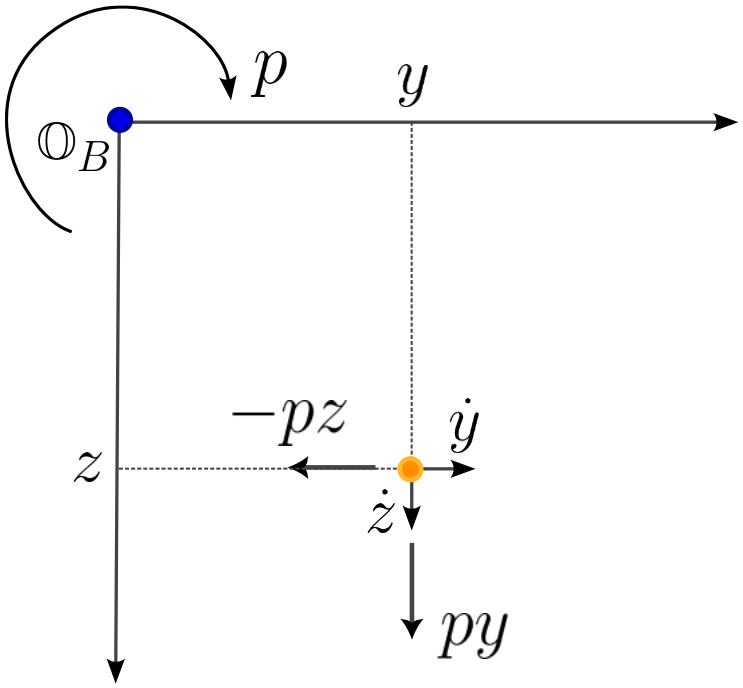
View looking along y axis

Generalised inertial acceleration

The tangential velocity equation "$v = \omega r$" is then applied multiple times using the rotation rate and radius for each isolated axis.

Here, a tangential velocity in the positive z direction is induced due to the radial offset y and rotation rate p, hence the py term





View looking along x axis

Generalised inertial acceleration

All of the velocities acting in each axis are summed,

$$u = \dot{x} + qz - ry$$
$$v = \dot{y} + rx - pz$$
$$w = \dot{y} + py - qx$$

Assuming rigid body, therefore no relative motion within body frame $\dot{x} = \dot{y} = \dot{z} = 0$

$$u = qz - ry$$
$$v = rx - pz$$
$$w = py - qx$$

Remember this is still the point velocity relative to the **body frame**

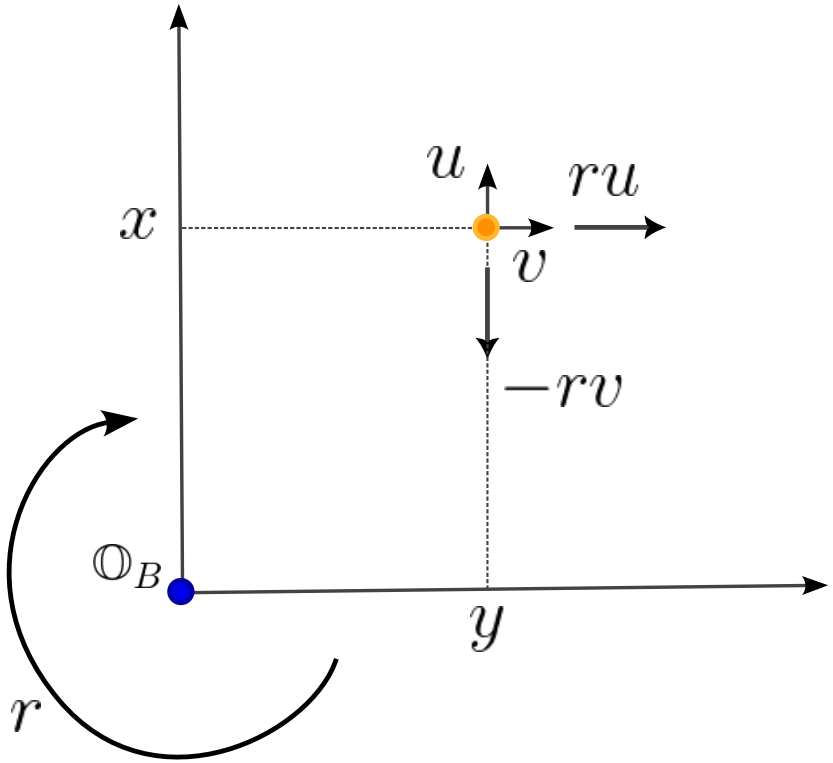
Generalised inertial acceleration

- The same process can be applied to tangential accelerations

$$a_x = \dot{u}_p - rv + qw$$
$$a_y = \dot{v}_p - pw + ru$$
$$a_z = \dot{w}_p - qu + pv$$

- Tangential accelerations arise due to mutual interaction of the linear velocity with the components of angular velocity

View looking along z axis



(other axes constructed similarly)

Generalised inertial acceleration

- Superimpose the local velocity components of a point with the velocity components of the c.g. The prime $(\cdot)'$ notation used to indicate overall motion of a point

$$u' = U + u = U - ry + qz$$
$$v' = V + v = V - pz + rx$$
$$w' = W + w = W - qx + py$$

- Differentiate with respect to time (using sum rule and partial derivatives)

$$\dot{u}' = \dot{U} - \dot{r}y - r\dot{y} + \dot{q}z + q\dot{z}$$
$$\dot{v}' = \dot{V} - \dot{p}z - p\dot{z} + \dot{r}x + r\dot{x}$$
$$\dot{w}' = \dot{W} - \dot{q}x - q\dot{x} + \dot{p}y + p\dot{y}$$

- Remembering $\dot{x}_p = \dot{y}_p = \dot{z}_p = 0$, hence

$$\dot{u}' = \dot{U} - \dot{r}y + \dot{q}z$$
$$\dot{v}' = \dot{V} - \dot{p}z + \dot{r}x$$
$$\dot{w}' = \dot{W} - \dot{q}x + \dot{p}y$$

Generalised inertial acceleration

- Superimpose the local acceleration components for a point with the acceleration components of the c.g.

$$a'_x = \dot{u}' - rv' + qw'$$
$$a'_y = \dot{v}' - pw' + ru'$$
$$a'_z = \dot{w}' - qu' + pv'$$

- Substitute previous equation for velocity of a point

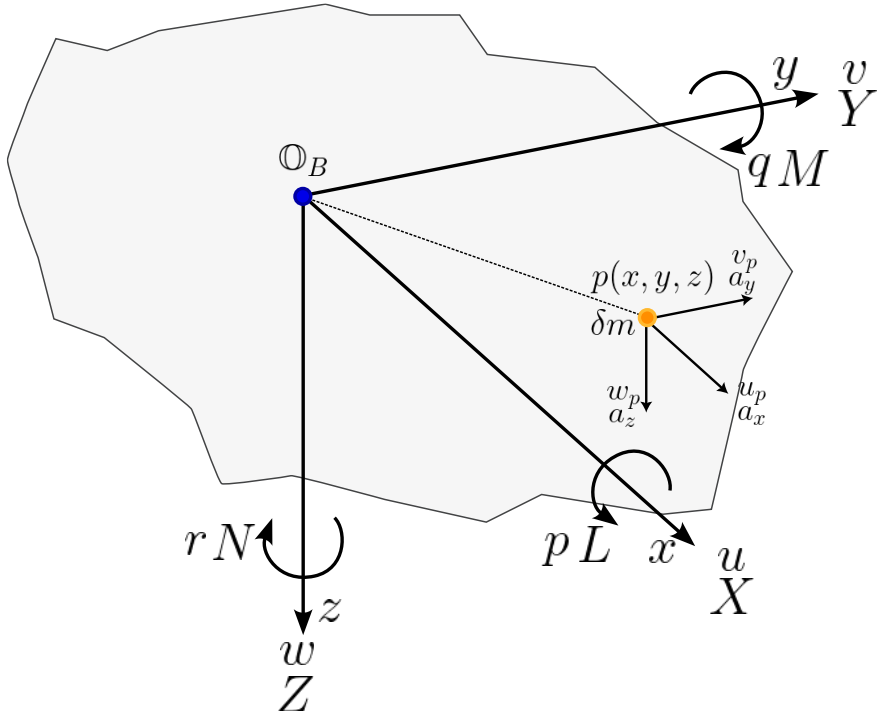
$$\dot{u}_p = \dot{u}_B - \dot{r}y_p - \dot{q}z_p$$
$$\dot{v}_p = \dot{v}_B - \dot{p}z_p - \dot{r}x_p$$
$$\dot{w}_p = \dot{w}_B - \dot{q}x_p - \dot{p}y_p$$

$$a'_x = \dot{U} - rV + qW - x(q^2 + r^2) + y(pq - \dot{r}) + z(pr + \dot{q})$$
$$a'_y = \dot{V} - pW + rU - x(pq + \dot{r}) + y(p^2 + r^2) + z(qr + \dot{p})$$
$$a'_z = \dot{W} - qU + pV + x(pr - \dot{q}) + y(qr + \dot{p}) - z(p^2 + q^2)$$

Generalised force equations

Assume each point p has an incremental mass δm

Sum of all accelerations acting on all incremental masses is the total force.



$$F = ma$$

Applied incrementally

$$\sum \delta m a'_x = X$$

$$\sum \delta m a'_y = Y$$

$$\sum \delta m a'_z = Z$$

Generalised force equations

Assume each point p has an incremental mass δm

Sum of all accelerations acting on all incremental masses is the total force. Example worked for X axis here

$$\sum \delta m a'_x = X$$

$$a'_x = \dot{U} - rV + qW - x(q^2 + r^2) + y(pq - \dot{r}) + z(pr + \dot{q})$$

$$X = \sum \delta m \dot{U} - \sum \delta m r V + \sum \delta m q W - \sum \delta m x (q^2 + r^2) + \sum \delta m y (pq - \dot{r}) + \sum \delta m z (pr + \dot{q})$$

at c.g. $\sum \delta m x = \sum \delta m y = \sum \delta m z = 0$

If not assuming forces, moments, velocities, accelerations at c.g. these terms must be kept

Generalised force equations

Remove zero terms and simplify

$$X = m(\dot{U} - rV + qW)$$
$$Y = m(\dot{V} - pW + rU)$$
$$Z = m(\dot{W} - qU + pV)$$

Finally, this can be written in vector notation,

$$\bar{F} = m\left[\dot{\bar{V}} + (\Omega \times \bar{V})\right]$$

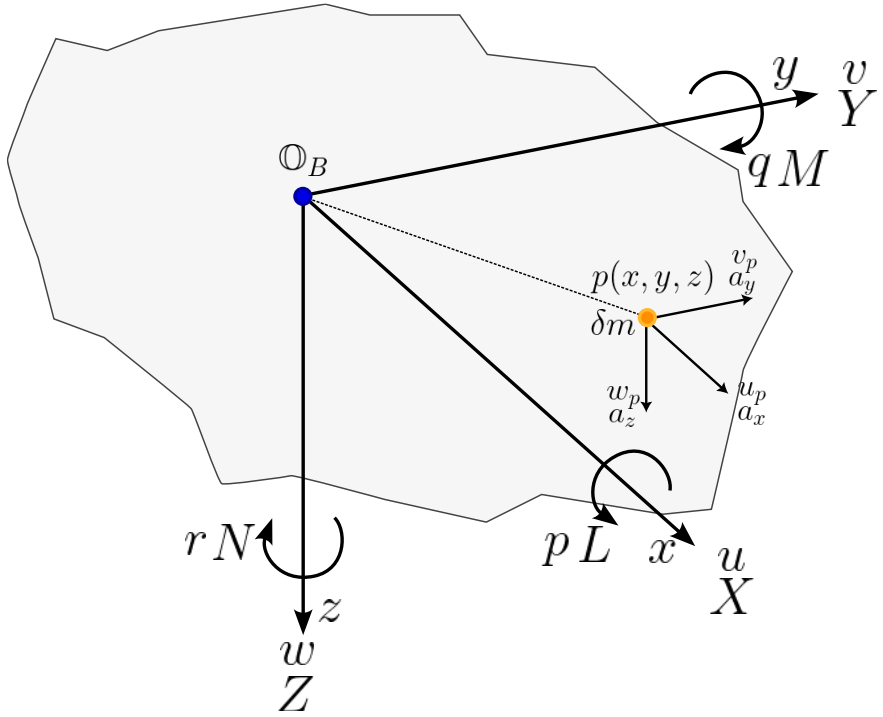
This is the generalised rigid body dynamics version of Newtons 2nd law applied to translational motion.

Generalised moment equations

Assume each point p has an incremental mass δm

Sum of all rotational accelerations acting on all incremental masses about the c.g. is the total moment.

$$T = I\dot{\omega}$$



Applied incrementally

$$\sum \delta m \ddot{\omega}_x = L$$

$$\sum \delta m \ddot{\omega}_y = M$$

$$\sum \delta m \ddot{\omega}_z = N$$

Generalised moment equations

Assume each point p has an incremental mass δm

Sum of all rotational accelerations acting on all incremental masses at known distance from centre of rotation. Example worked for X axis here

$$\sum \delta m(ya'_z - za'_y) = L$$

$$a'_y = \dot{V} - pW + rU - x(pq + \dot{r}) + y(p^2 + r^2) + z(qr + \dot{p})$$
$$a'_z = \dot{W} - qU + pV + x(pr - \dot{q}) + y(qr + \dot{p}) - z(p^2 + q^2)$$

$$L = \sum \delta my\dot{V} - \sum \delta mypW + \sum \delta myrU - \sum \delta myx(pq + \dot{r}) + \sum \delta my^2(p^2 + r^2) + \sum \delta myz(qr + \dot{p}).....$$

Dimension of mass x distance^2

at c.g. $\sum \delta mx = \sum \delta my = \sum \delta mz = 0$

Generalised moment equations

- After expanding and factoring terms involving mass and position

$$L = \dot{p}\sum \delta m(y^2+z^2) + qr\sum \delta m(y^2-z^2) + (r^2-q^2)\sum \delta myz - (pq+\dot{r})\sum \delta mxz + (pr-\dot{q})\sum \delta mxy$$

- Any term which is the sum of mass x distance^2 is an inertia I

$$L = I_x\dot{p} - (I_y - I_z)qr + I_{xy}(pr - \dot{q}) - I_{xz}(pq + \dot{r}) + I_{yz}(r^2 - q^2)$$

Generalised force equations

Remove zero terms and simplify

$$L = I_x\dot{p} - (I_y - I_z)qr + I_{xy}(pr - \dot{q}) - I_{xz}(pq + \dot{r}) + I_{yz}(r^2 - q^2)$$
$$M = I_y\dot{q} + (I_x - I_z)pr - I_{xy}(qr - \dot{p}) + I_{xz}(p^2 - r^2) + I_{yz}(pq - \dot{r})$$
$$N = I_z\dot{r} - (I_y - I_y)pq + I_{xy}(q^2 - p^2) + I_{xz}(qr + -dotp) + I_{yz}(pr - \dot{q})$$

Finally, this can be written in vector notation,

$$\bar{M} = I\dot{\Omega} + (\Omega \times (I\Omega))$$

Where I is a 3x3 inertia matrix $I = \begin{bmatrix} I_x & I_{xy} & I_{xz} \\ I_{xy} & I_y & I_{yz} \\ I_{xz} & I_{yz} & I_z \end{bmatrix}$

This is the rigid body dynamics version of Newtons 2nd law applied to rotational motion.

Conclusions

- We have formally linked forces and moments to accelerations in the body frame
- Both translational and rotational
- Origin of all references frames is centre of mass
- Mass and distribution of mass remain constant
- No other assumptions so far